



TMRC 2026

(13th) Conference Technology Survey

August 2026, UCSD, San Diego ,CA.



Survey this year 1/2

Survey of opinions on technology intercepts for HDD and MRAM industry.

* 1. Describe your affiliation ?

- ☐ HDD Industry Member
- ☐ MRAM Industry Member
- ☐ Academia
- ☐ Vendor
- ☐ Other

Survey issued continuously over the meeting period.
Differs from pre/post survey past years.

•The response rate was lower this year, so we consolidated the pre and post conference survey into one.

* 2. What is the Maximum Areal Density Capability expected for Perpendicular/Shingled/Two dimensional - magnetic recording extensions?

3. What is the expected Year of Technology introduction to HDD Products ?

	2026	2027	2028	2029	2031	2033	2035	Never
BPM	☐	☐	☐	☐	☐	☐	☐	☐
HAMR	☐	☐	☐	☐	☐	☐	☐	☐
MAMR	☐	☐	☐	☐	☐	☐	☐	☐
HDMR(BPM+HAMR)	☐	☐	☐	☐	☐	☐	☐	☐
Multilayer Magnetic Recording	☐	☐	☐	☐	☐	☐	☐	☐
All Optical Recording	☐	☐	☐	☐	☐	☐	☐	☐
2 Heads per surface Write and Readback	☐	☐	☐	☐	☐	☐	☐	☐
Non local Spin valve or Graphene	☐	☐	☐	☐	☐	☐	☐	☐

New 2026

Survey this year 2/2

3

4. Likely intercept of Rotating Storage as Practical Bulk Archive medium

2026	2027	2029	2031	2033	Never
☐	☐	☐	☐	☐	☐

5. Timing for the product delivery of 3D DRAM (Not HBM -3D stacked DRAM)

2026	2027	2029	2031	2033	Never
☐	☐	☐	☐	☐	☐

6. What are the practical limits of the maximum number of NAND layers

- ☐ 200
- ☐ 500
- ☐ 1000
- ☐ Don't know

7. Which emerging memory will become the Future Storage Class Memory?

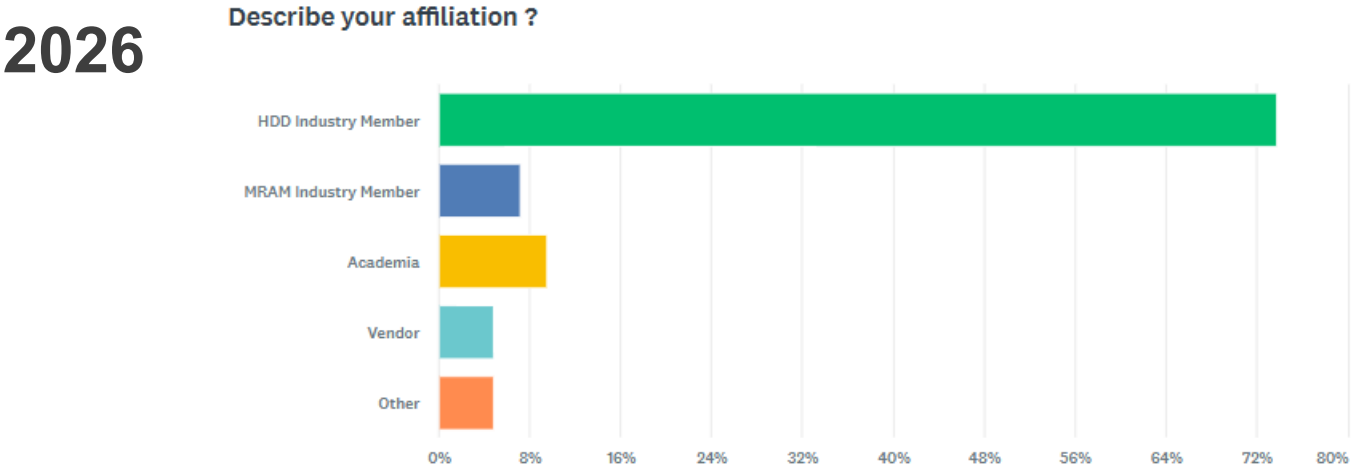
- ☐ FRAM
- ☐ ReRAM
- ☐ MRAM
- ☐ Other

Population of respondents up to 08/04/26 (during conference)

As with 2015-25.

Dominant responses from HDD members.

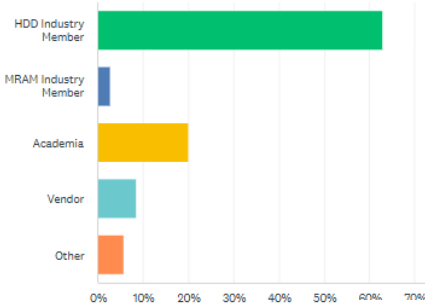
Improved MRAM industry response so far this year.



2025

Describe your affiliation ?

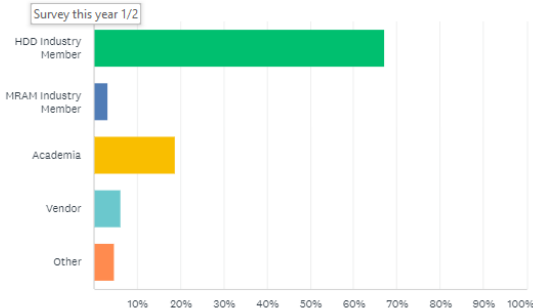
Answered: 35 Skipped: 0



2024

Describe your affiliation ?

Answered: 64 Skipped: 0

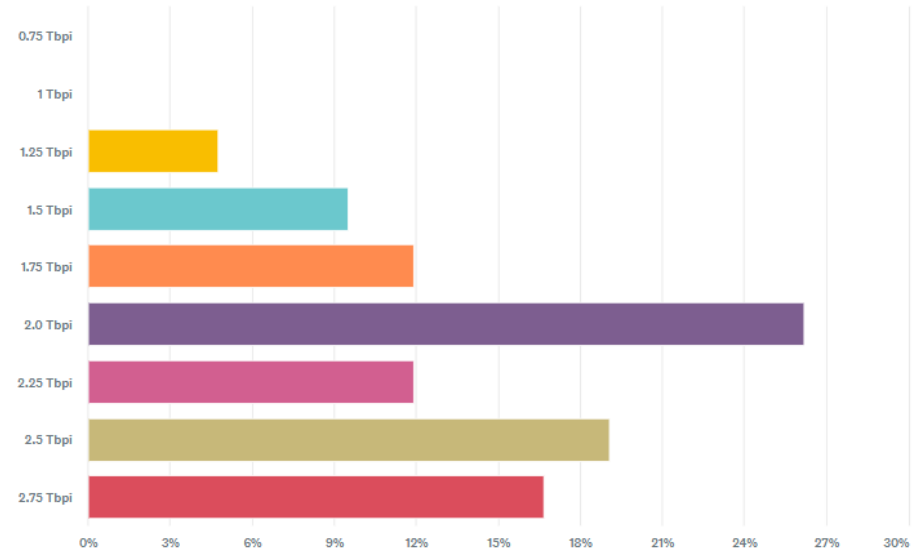


Maximum ADC, for conventional technology

- Increased optimism for “conventional” technology. (is HAMR and ePMR now “conventional”?)
- Modes at 2 and 2.5 Tbps
- Pattern very similar to 2018-2024, being weakly bimodal now.

2026

What is the Maximum Areal Density Capability expected for Perpendicular/Shingled/Two dimensional - magnetic recording extensions?



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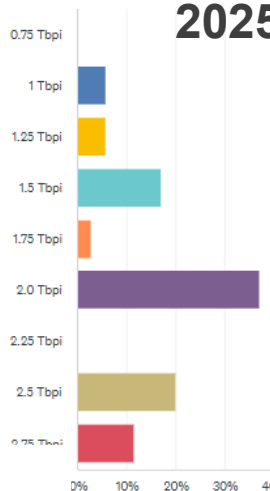
Answered: 35 Skipped: 0

Answered: 64 Skipped: 0

Industry

Academia

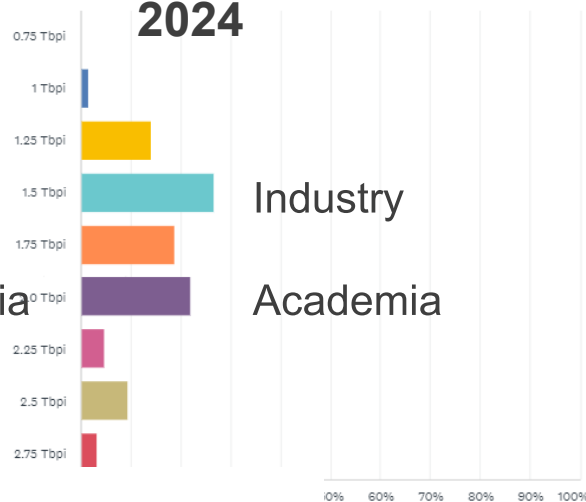
2025



Industry

Academia

2024

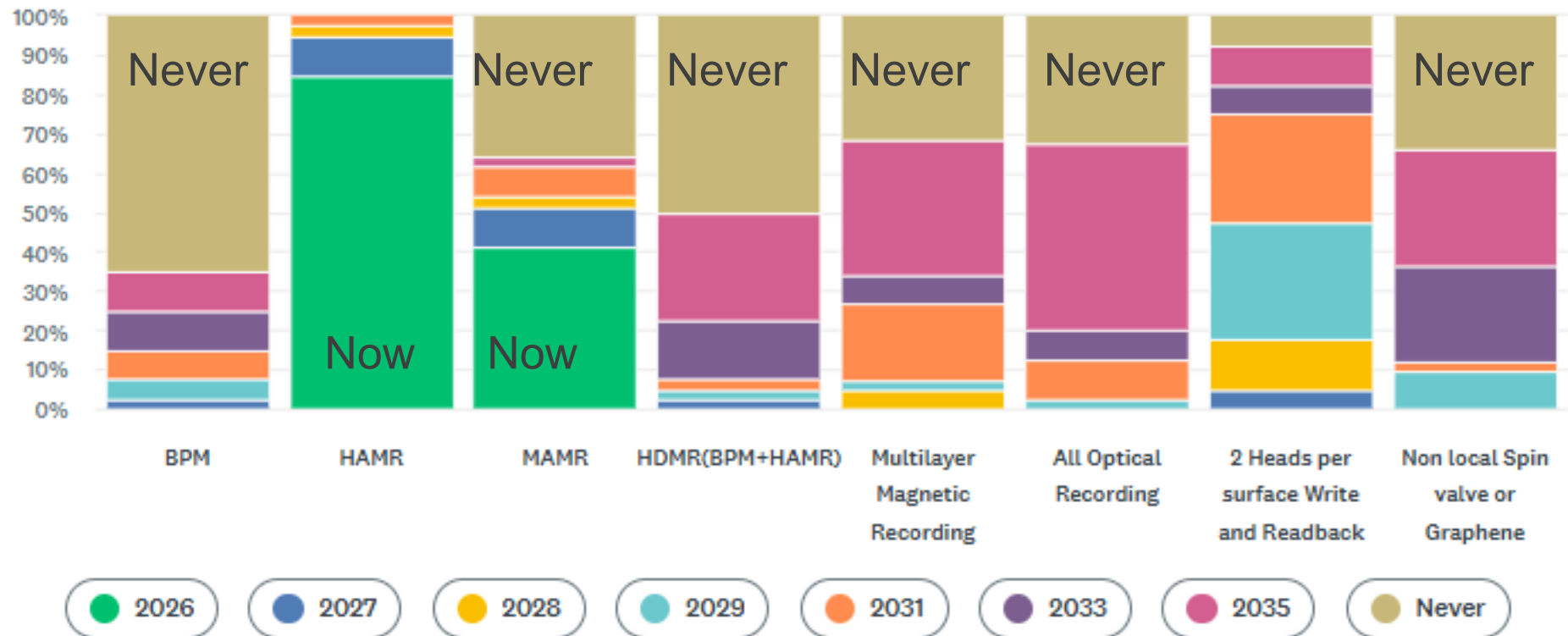


Industry

Academia

Expected introduction year 2026 survey

What is the expected Year of Technology introduction to HDD Products ?



Expected introduction year

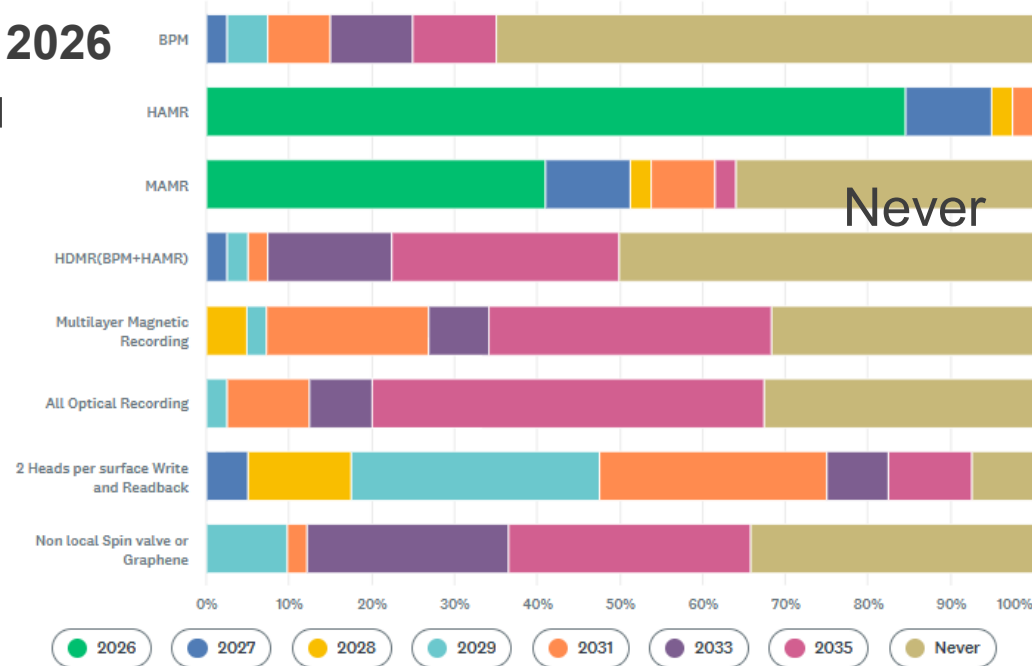
Pessimism for MAMR reduced in 2017, and improved 2018- drift back up 2019+ 2023 increased significantly- and stayed.

BPM/Heated Dot remains pessimistic

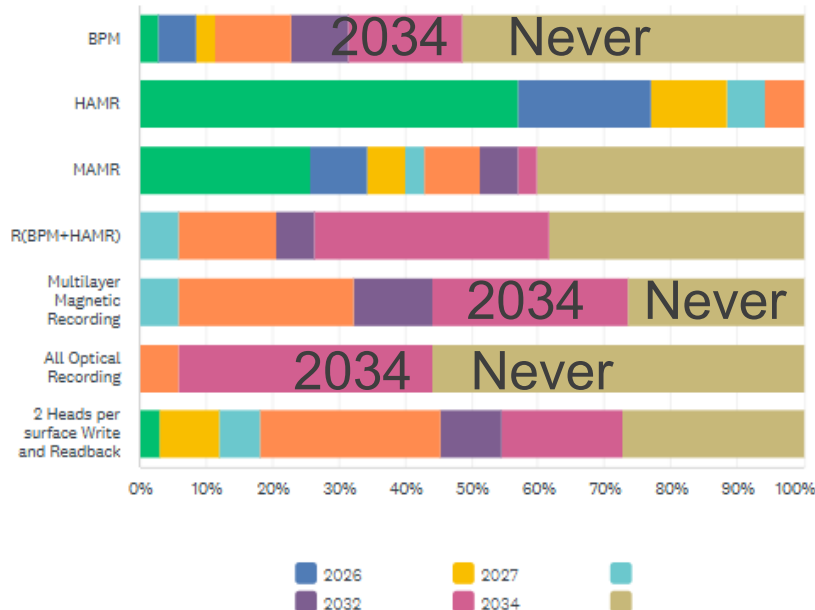
The last 5 years- MAMR and HAMR – is “soon”/”Now”.

Multilayer and all optical starts out as very immature and not ready.

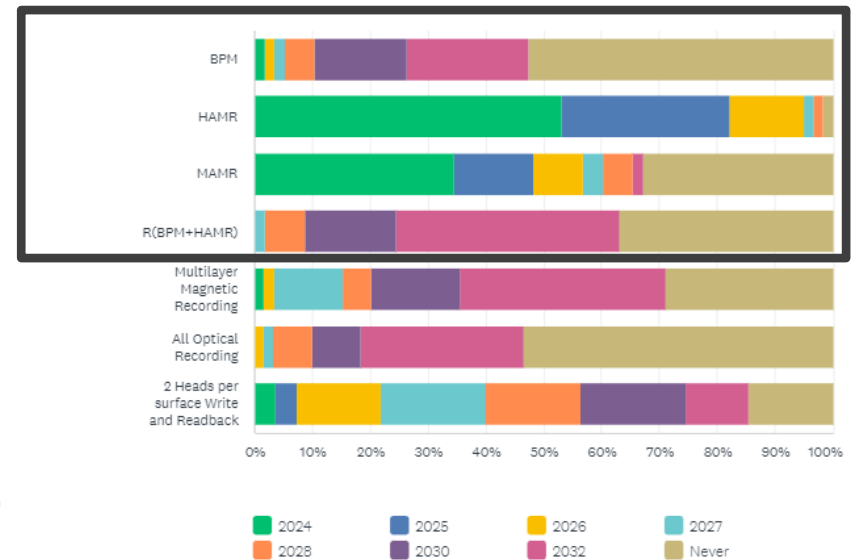
2026



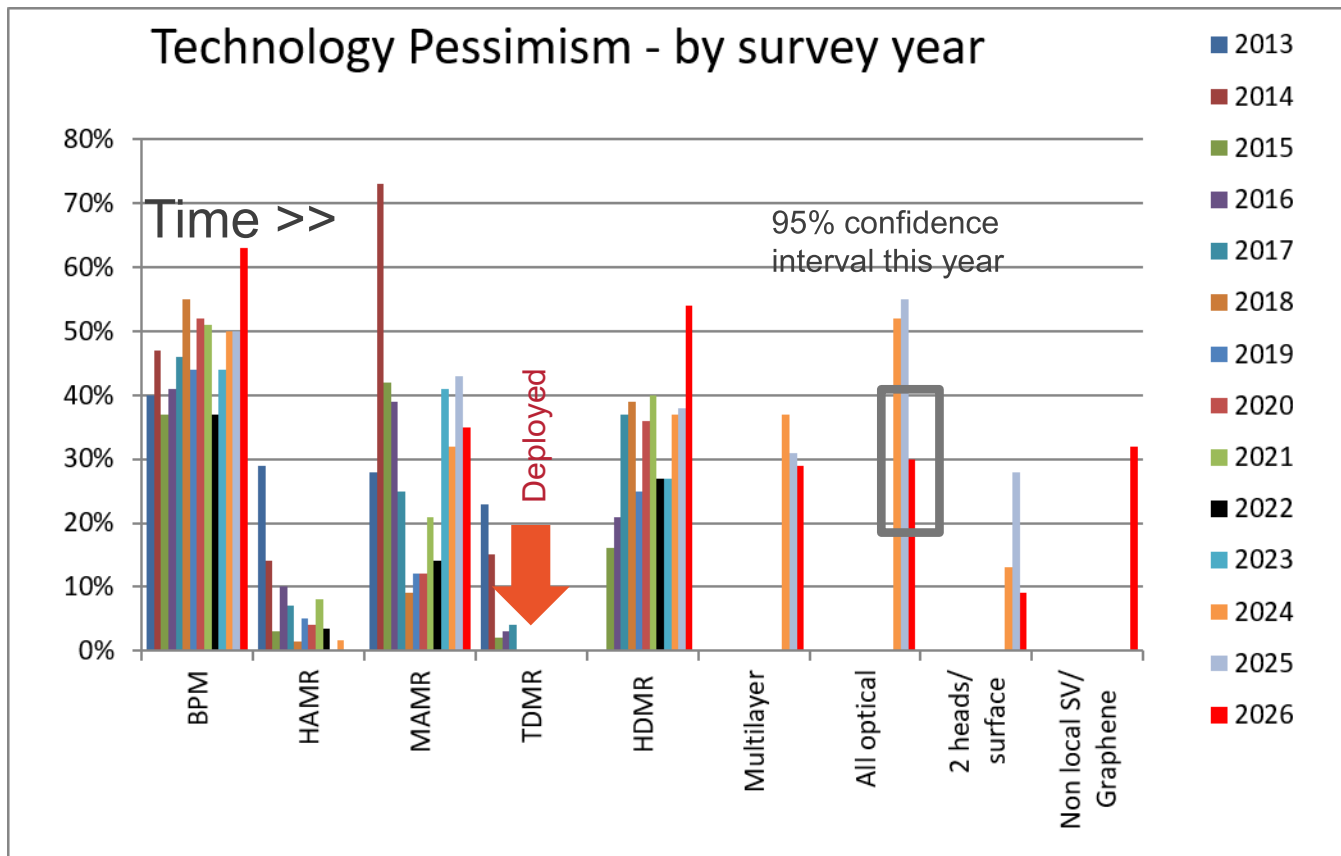
2025



2024



Technology pessimism(Never): Compare 2026 with 2025-2013



From left to right..

- BPM appears stable and poor.
- HAMR confidence – Zero Pessimism for last few years. Suggest we remove next year.
- MAMR hit a bad patch 2014, started recovery in 2016-2017, and significantly improved 2018-but poor 23/24/25/26
- TDMR Launched 2017 into product – so removed 2018.
- HDMR confidence – better than BPM but still poor.
- 2 heads per surface does not appear a discovery issue- but a business choice.

Technology pessimism(Never): Compare 2026 with 2025-2013

Technology	BPM	HAMR	MAMR	TDMR	HDMR	Multilayer	All optical	2 heads/ surface	Non local SV/ Graphene
2013	40%	29%	28%	23%					
2014	47%	14%	73%	15%					
2015	37%	3%	42%	2%	16%				
2016	41%	10%	39%	3%	21%				
2017	46%	7%	25%	4%	37%				
2018	55%	2%	9%		39%				
2019	44%	5%	12%		25%				
2020	52%	4%	12%		36%				
2021	51%	8%	21%		40%				
2022	37%	4%	14%		27%				
2023	44%	0%	41%		27%				
2024	50%	2%	32%		37%	37%	52%	13%	
2025	50%	0%	43%		38%	31%	55%	28%	
2026	63%	0%	35%		54%	29%	30%	9%	32%

Year we surveyed you

From left to right..

- BPM appears stable and poor.
- HAMR confidence – Zero Pessimism for last few years. Suggest we remove next year.
- MAMR hit a bad patch 2014, started recovery in 2016-2017, and significantly improved 2018-but poor 23/24/25/26
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Technology Introduction Year

Technology	BPM	HAMR	MAMR	TDMR	HDMR	Multilayer	All optical	2 heads/ surface	Non local SV/ Graphene
2013	2022	2018	2018	2017	N/A				
2014	2022	2018	2018	2017	N/A		*Pessimism is high So confidence on introduction year is poor.		
2015	2023*	2018	2020*	2017	2022				
2016	2024*	2018	2020*	2017	2023				
2017	2024*	2019	2021*	2018	2023*				
2018	2029*	2020	2020	-	2029*				
2019	2029*	2020	2020	-	2029*				
2020	2030*	2021	2021	-	2030*				
2021	2030*	2023	2022	-	2030*				
2022	2030*	2024	2022	-	2030*				
2023	2030*	2023	2023	-	2030*				
2024	2030*	2025	2025	-	2031*	2030*	2030	2028	
2025	2034*	2025	2025*	-	2034*	2032*	2034*	2032*	
2026	2035*	2026	2026*	-	2035*	2035*	2035*	2028	2035*

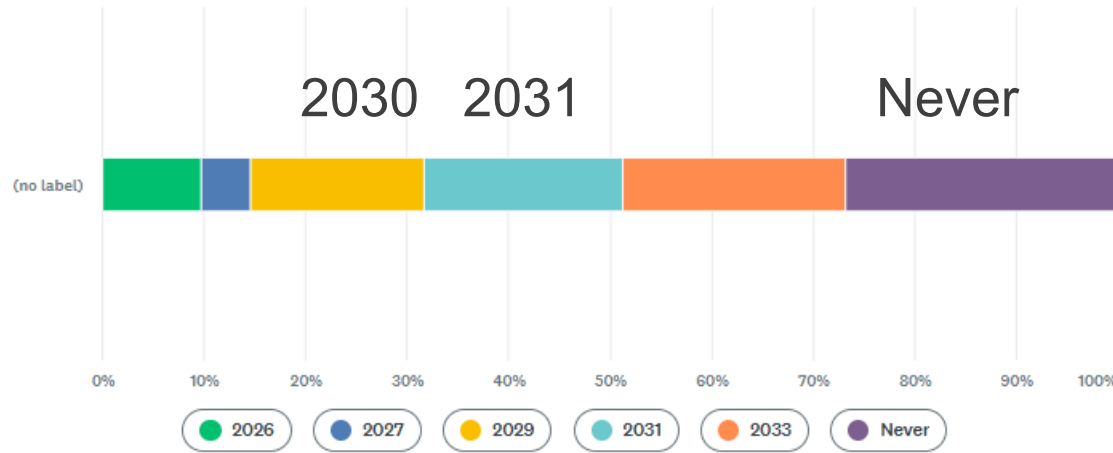
TDMR released in 2017 and removed as a question.

HAMR is locked in on the earliest date in the survey.

BP/HDMR/multilayer/All optical/ NLSV/Graphene are all in the –discovery mode.

Will HDD take over from Tape?

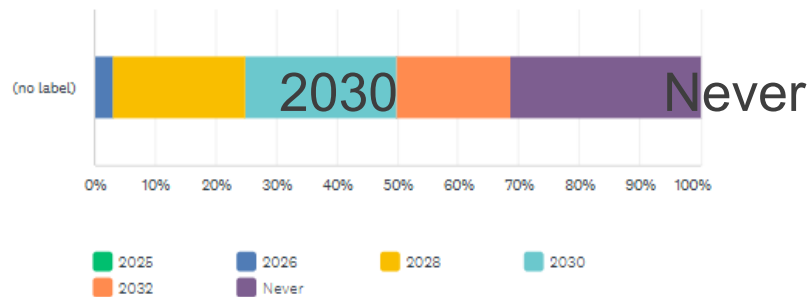
Likely intercept of Rotating Storage as Practical Bulk Archive medium



2025 Survey

Likely intercept of Rotating Storage as Practical Bulk Archive medium

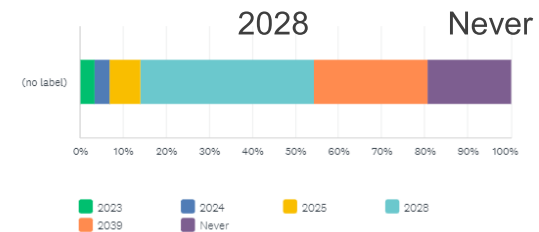
Answered: 32 Skipped: 3



2024 Survey

Likely intercept of Rotating Storage as Practical Bulk Archive medium

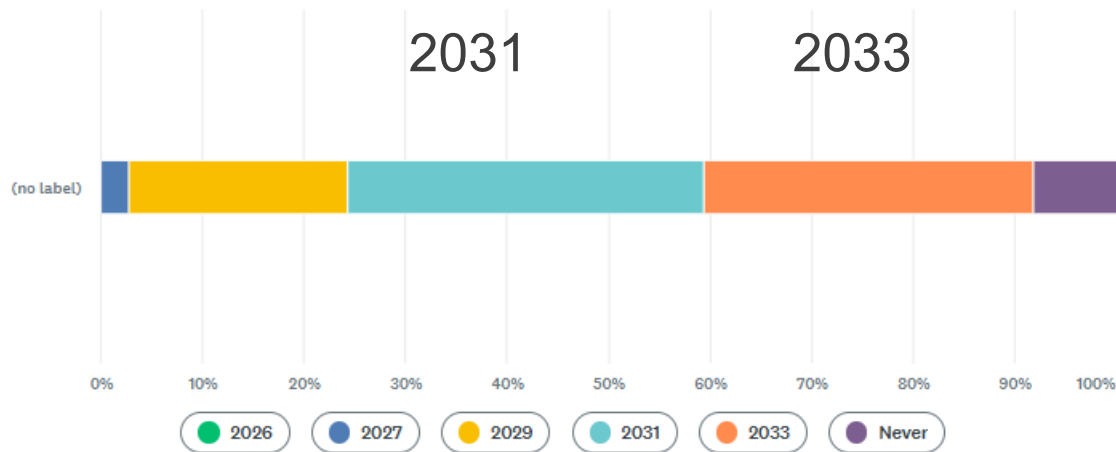
Answered: 57 Skipped: 7



	2023	2024	2025	2028	2039	NEVER	TOTAL	WEIGHTED AVERAGE
(no label)	3.51%	3.51%	7.02%	40.35%	26.32%	19.30%	57	4.40
	2	2	4	23	15	11		

3D DRAM- when?

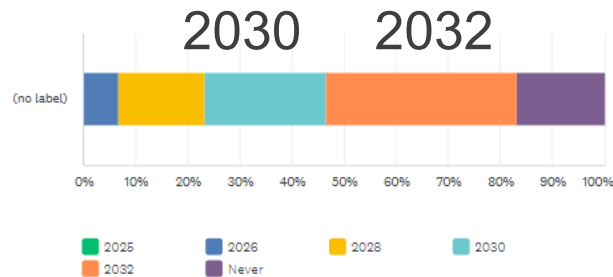
Timing for the product delivery of 3D DRAM (Not HBM -3D stacked DRAM)



2025 Survey

Timing for the product delivery of 3D DRAM

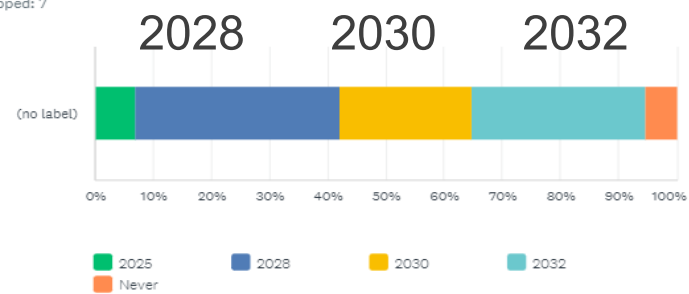
Answered: 30 Skipped: 5



2024 Survey

Timing for the product delivery of 3D DRAM

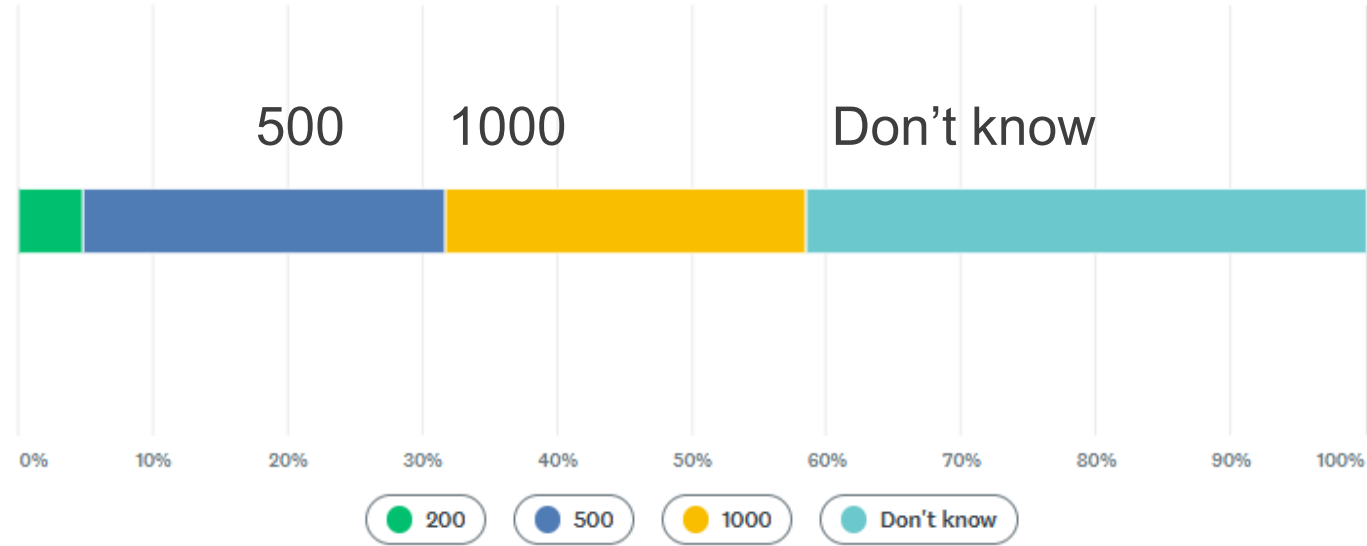
Answered: 57 Skipped: 7



	2025	2028	2030	2032	NEVER	TOTAL	WEIGHTED AVERAGE
(no label)	7.02%	35.09%	22.81%	29.82%	5.26%	57	2.91
	4	20	13	17	3		

NAND Question (the competing pressure).

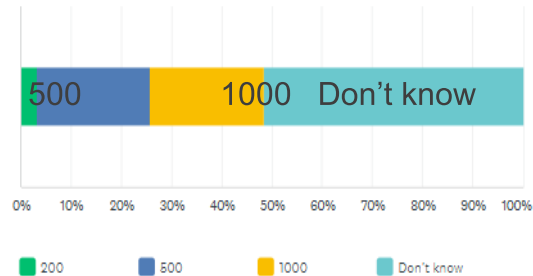
What are the practical limits of the maximum number of NAND layers



2025

What are the practical limits of the maximum number of NAND layers

Answered: 31 Skipped: 4



2024

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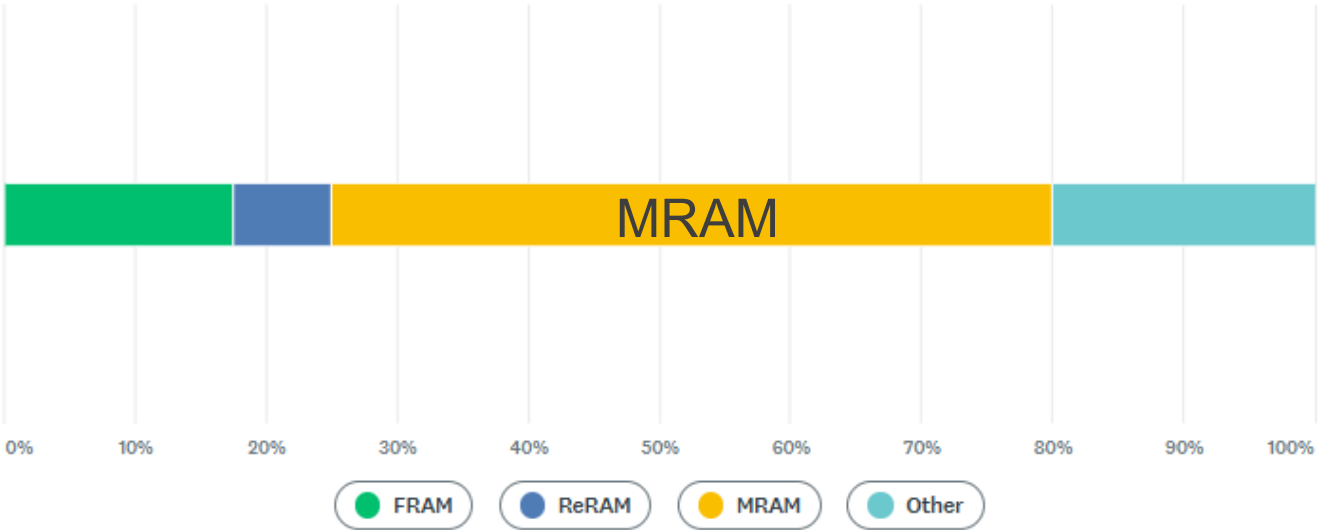
Answered: 62 Skipped: 2



ANSWER CHOICES	RESPONSES	
▼ 200	9.68%	6
▼ 500	29.03%	18
▼ 1000	16.13%	10
▼ Don't know	45.16%	28
TOTAL		62

New Solid state Technologies.

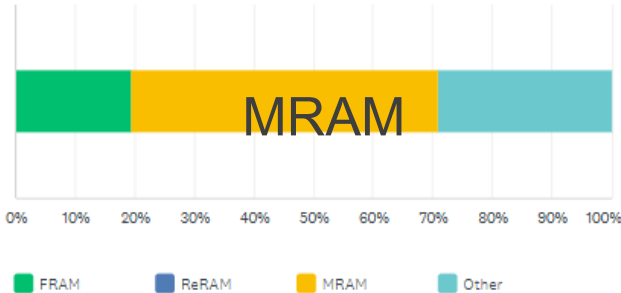
Which emerging memory will become the Future Storage Class Memory?



2025

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Answered: 31 Skipped: 4



2024

Which emerging memory will become the Future Storage Class Memory?

Answered: 57 Skipped: 7



ANSWER CHOICES	RESPONSES	
FRAM	12.28%	7
ReRAM	10.53%	6
MRAM	57.89%	33
Other	19.30%	11
TOTAL		57



THANK YOU

